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A Simple Model to Assess Nitrogen and Phosphorus Contamination in Ungauged Surface Drainage Networks: Application to the Massaciuccoli Lake Catchment, Italy

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Modeling is a common practice to evaluate factors affecting water quality in environmental systems impaired by point and nonpoint losses of N and P. Nevertheless, in situations with inadequate information, such as ungauged basins, a balance between model complexity and data availability is necessary. In this paper, we applied a simplified analytical model to an artificially drained floodplain in central-western Italy to evaluate the importance of different nutrient sources and in-stream retention processes and to identify critical source areas. We first considered only a set of chemical concentrations in water measured from February through May 2008 and from November 2008 through February 2009. We then broadened available data to include water discharge and hydraulic-head measurements to construct a hydrogeological model using MODFLOW-2000 and to evaluate the reliability of the simplified method. The simplified model provided acceptable estimates of discharge (ranging from 0.03–0.75 $m^3 s^{-1}$) and diffuse nutrient inputs from water table discharge and in-stream retention phenomena. Estimates of PO_4–P and total P retention (ranging from 1.0 to 0.6 $\mu g\ m^{-2}\ s^{-1}$ and from 1.18 to 0.95 $\mu g\ m^{-2}\ s^{-1}$ for PO_4–P and total P, respectively) were consistent with the range of variability in literature data. In contrast, the higher temporal variability of nitrate concentrations decreased model accuracy, suggesting the need for more intensive monitoring. The model also separated the dynamics of different reaches of the drainage network and identified zones considered critical source areas and buffer zones where pollutant transport is reduced.

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In the context of environmental systems impaired by losses of N and P, planners require assistance evaluating the contribution of point- and nonpoint-source losses (Bowes et al., 2008) and identifying critical source areas (i.e., areas where a significant source of nutrient input is directly connected to receiving waters) (Heathwaite et al., 2005). Such evaluations of land-use effects on water quality are not straightforward tasks considering the many factors that can affect the mobilization and transport of pollutants, such as land use, agricultural management practices, and soil type. Spatial and temporal changes in driving variables, system properties, and their interactions generate a complexity that makes description of all system dimensions impossible. In comparison, the information that one can acquire generally is limited to simple and inexpensive data, such as pollutant chemical concentrations in surface waters. To circumvent these difficulties, modeling is a common solution because it simplifies system description and reduces complexity (De Wit and Pebesma, 2001).

Several modeling approaches for representing the functioning of surface drainage networks exist, depending on the extent of available information. Physically based models integrate multiple scales of system complexity, such as the 1D models DRAINMOD (Campbell et al., 1995; Fernandez et al., 2006) and OTIS (Bencala and Walters, 1983) or the 2D/3D models HYDRUS (van Beek et al., 2007) and MODFLOW (Harbaugh et al., 2000). This also can be the case with empirical/analytical approaches, which evaluate pollutant fluxes among several compartments of a studied system (e.g., Seneque/Rivestrahler model [Billen et al., 2007; Thieu et al., 2009] and INCA [Whitehead et al., 1998]). In contrast, simpler aggregated models require much less data to consider elementary stores and the fluxes between them (e.g., PolFlow [de Wit and Bendoricchio, 2000; de Wit and Pebesma, 2001]; two-store model [Ruiz et al., 2002]). Other approaches are based solely on statistical analysis of available data that describe the entire system or its subcompartments (e.g., Rupp et al., 2004; van Beek et al., 2004; Chardon and Schoumans, 2007). Statistical

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Abbreviations: RE, relative error; WWTP, wastewater treatment plant.

methods were not conceived to locate critical sources, however, and their ability to predict reactions to changes in systems other than those for which they were developed is uncertain.

Developing models that describe the dynamics of surface drainage networks is challenging, considering the intermittent character of water regimes, ditch interconnections, and, in some circumstances, changes in water flow direction and intensity when drainage is artificially controlled by pumping stations (e.g., reclamation catchments). To our knowledge, modeling drainage-network dynamics remains unexplored, especially with limited information. This paper examines this issue and describes the development and evaluation of a simplified analytical model whose physically based parameters are estimated by inverse evaluation. We focus on nonlinear processes of pollutant mobilization and retention in ditch networks, which are considered to be of major importance to control the environmental fate of pollutants (Ruiz et al., 2002; van Beek et al., 2004). The model was applied to a typical artificially drained coastal floodplain, the Lake Massaciuccoli catchment in central-western Italy, using a limited database and restricted knowledge of chemical concentrations in water. We aimed to develop, in the context of low data availability, a descriptive tool of the system controlling nutrient and pollutant fluxes between land, groundwater, and drainage reaches to evaluate critical sources for pollution and associated mitigation options. We evaluated the simplified model by comparing its predictions to observed data and MODFLOW-2000 predictions of drainage-network functioning from a dataset that encompasses water discharge, field velocity, and hydraulic-head measurements.

Materials and Methods

Study Site

Lake Massaciuccoli and the nearby palustrine areas represent one of the most important residual coastal marshy areas in Tuscany (Italy). Since the 1970s, the lake, initially oligotrophic (Brunelli and Cannicci, 1942), progressively converted to a eutrophic/hypereutrophic state (Cenni, 1997). Since 1930, most of the plain has been drained by a complex network of artificial canals and ditches and pumping stations, forcing water from reclaimed areas into the lake, while the natural network is poorly developed.

The study area (2255 ha) is located in the southern part of the lake catchment and consists of two mechanically drained ungauged subcatchments, the Vecchiano and Massaciuccoli subcatchments (Fig. 1a). In terms of hydrological boundaries, the domain is bordered by carbonate and arenaceous reliefs on the east and by a potentiometric high in the sandy coastal shallow aquifer on the west (Fig. 1b). South and north of the area, the Serchio River and Lake Massaciuccoli, respectively, are hydraulically connected to the superficial phreatic aquifer (Rossetto et al., 2010b). This aquifer consists of sand deposits (30–40 m thick), which are overlain by recent deposits that range from silty to clayey sands to peaty clay and then to peat as one moves northward. The mean annual rainfall is around 900 mm, with two typical peaks in autumn and spring. The artificial network drains the superficial aquifer and some excess rainfall, maintaining a water table depth suitable for cultivation. During summer (July–October), it supplies irrigation water from the lake to cropland by inverting water flow direction. The water table depth ranges

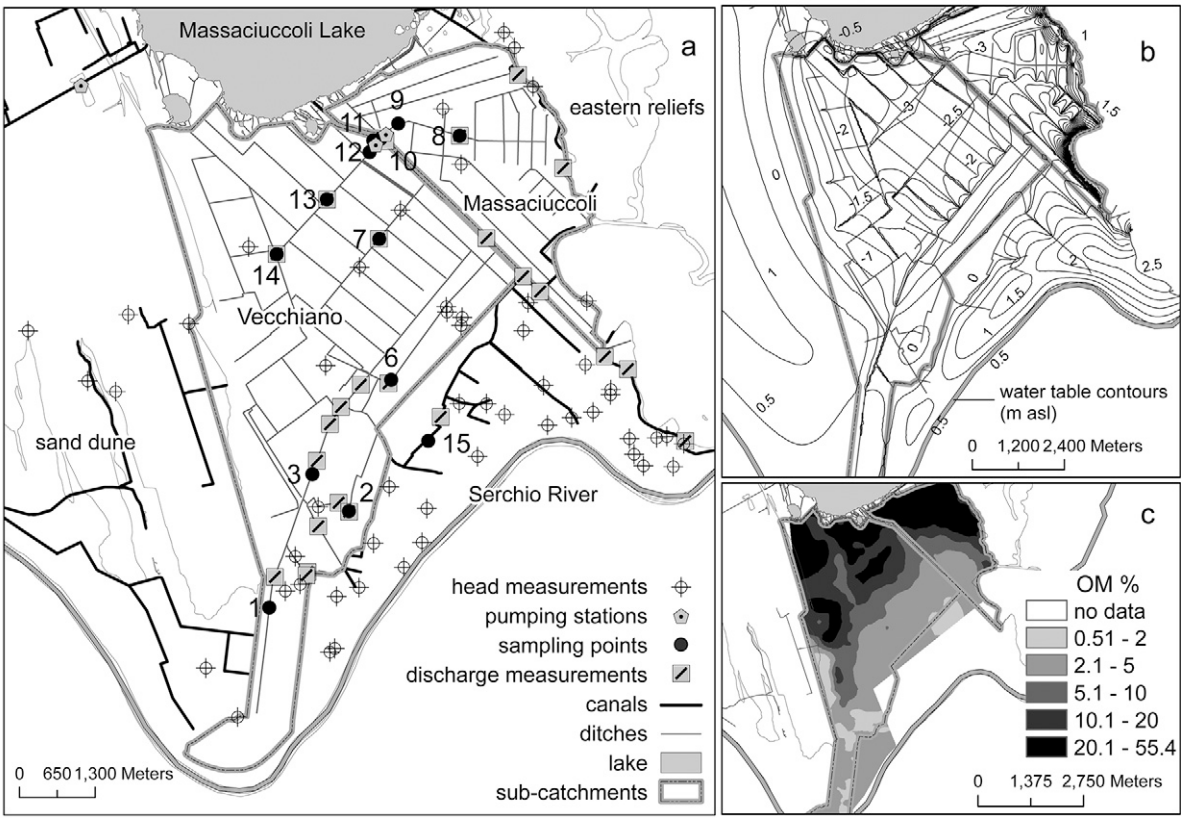


Fig. 1. Maps of the study area showing (a) the ditch network and water sampling, discharge, and head-measurement locations. (b) Soil organic matter (OM) content (%). (c) Piezometric levels (m above sea level).

from 0.5 to 2 m as one moves from the pumping stations toward the southern boundary of the domain.

At the northern end of the two subcatchments, peat soil (eutri-sapric and endo-salic histosols; Accolti Gil, unpublished data) organic matter reaches values up to 55% (Silvestri et al., 2002). Moving southward, clay loam, loam, and silt loam soils prevail, with progressively decreasing organic matter content (Fig. 1c). These soils are classified as calcari-fluvic cambisols (Accolti Gil, unpublished data). After land reclamation started, the peatland subsided (2–3 m in 70 yr), leaving the lake perched above the lower drained area, now 0 to 3 m below mean sea level (Autorità di Bacino del Fiume Serchio, unpublished data). This also determines recharge from the lake to the superficial aquifer due to seepage, currently estimated at 25 to 30% of total annual recharge (Rossetto et al., 2010a). Land use is characterized by intensive agriculture (80% of the surface) and periurban infrastructures. The effluents of a 4250-person-equivalent wastewater treatment plant (WWTP) are discharged at point 2 (Fig. 1a). Cropping systems are based on spring crops, continuous maize (*Zea mays* L.), and sunflower (*Helianthus annuus* L.)–wheat (*Triticum* spp. L.) or maize–wheat rotations in the peatland and mainly winter cereals in the remaining part of the subcatchments. A small part of the Vecchiano, located along the southeastern border, is managed by smallholders or hobby farmers and is characterized by highly fragmented cropland: mixed woody, row, and horticultural crops (Silvestri et al., in press).

Sampling and Laboratory Analysis

Concentrations of Cl, Na, K, Mg, Ca, total phosphorus (TP), PO_4–P, and NO_3–N were measured monthly at 12 sampling locations (Fig. 1a) following a synoptic water-sampling procedure in two distinct monitoring periods, both during drainage conditions but with different cropping systems: February through May 2008 (hereafter P-I) and November 2008 through February 2009 (hereafter P-II). In our study, monthly sampling seemed a good compromise between an economically feasible and a sufficiently representative estimate of mean/median concentrations in small streams outside stormflow periods (Kronvang and Bruhn, 1996; Robertson, 2005).

To capture the spatial patterns of chemical inputs and outputs better, the sampling sites (Fig. 1a) were uniformly distributed along the drainage network depending on accessibility, presence of point sources, and ditch connections. Points 3 and 6 were located downstream of the WWTP. Points 1, 15, 7, 8, 13, and 14 drained mainly cropland, while points 12 and 9 were located at the outlets of the Vecchiano and Massaciuccoli subcatchments, respectively. Samples also were collected from the center of the lake.

Stream water samples were collected over 1 to 2 d, assuming that discharge and concentrations were stable during that time. At each site, water samples (5–6 L) were collected near the center of the stream at a depth of 0.2 to 0.3 m. At the same time, water and air temperature, pH, and electrical conductivity were measured using portable devices (XS pH 6 and XS COND 6, Oakton-Eutech Instruments). Alkalinity was measured directly in the field by acidimetric titration with 0.1 mol L^{-1} hydrochloric acid (HCl) using a portable micro-dosimeter. Each sample was divided into subsamples; the subsample for PO_4–P and NH_4–N analysis was filtered through 0.45-μm cellulose filters and collected in glass bottles, and the subsample for TP was acidified with 1:1 sulfuric acid (H_2SO_4). Subsamples for major anions and cations were put into PTE bottles. Subsamples were filtered and acidified with 1:1 HCl for Ca and Mg analysis; subsamples for Cl, SO_4, Na, and K were not filtered. All samples were stored in the dark at 4°C until analysis. Within 2 wk after sampling, samples were analyzed for concentrations of NO_3–N, K, Na, Ca, and Mg ions by ion chromatography (Dionex Dx-100 and Dx-500 ion chromatograph). The Cl^- concentration was detected with a potentiograph (Herisau E536, Metrom). The PO_4–P concentration was determined by UV/VIS spectrophotometry (PerkinElmer Lambda 1), as was TP after sample digestion with peroxidisulfate and H_2SO_4 at 120°C. Ammonium-N was detected by a selective electrode (Orion 95–12). All samples were checked for charge balance error (Freeze and Cherry, 1979), repeating the analysis until the charge balance error was ≤5%, if needed. As additional QA/QC procedures, blank samples were included in each analysis, as were replicates at different dilutions, to check for matrix effects. Standards also were included within the same batch.

Statistical analysis was performed using R statistical software (version 2.12.0, R Foundation for Statistical Computing, http://www.r-project.org). To calibrate and validate the MODFLOW-2000 groundwater model, an acoustic flow meter measured discharges from June 2008 to July 2009 (acoustic digital current meter, OTT Messtechnik GmbH). A groundwater head survey also occurred during the monitoring period (Fig. 1a).

Analytical Model Development

We considered the hydrological domain as a superficial phreatic aquifer drained by a dense ditch network. Following Boorman (2003), the drainage network was modeled as a series of successive reaches delineated by sampling points, each representing a storage element. For a given node k the flux F_k (mg s^{-1}) of a given solute (i.e., the mass passing through the node section per unit time) was calculated as the sum of the F_i fluxes of the connected upstream nodes:

$$F_k = \sum_i \alpha F_i \qquad [1]$$

where index i identifies the upstream nodes (e.g., for k = 3, i = 1,2,15; Fig. 2), and α is a repartition coefficient defining the apportionment of F_i when node i is connected to two or more downstream nodes (e.g., node 3 connected to nodes 13 and 6 in Fig. 2).

We introduced two new terms to represent the contribution of in-stream processes and diffuse inputs derived from water table discharge. In-stream processes include retention (e.g., precipitation, chemical transformations, sorption onto bed sediments, algal uptake) and release phenomena (e.g., bed erosion sediment resuspension, remineralization, desorption, and dissolution) that are difficult to quantify separately. We assumed the former processes prevail over the latter during base-flow conditions characterized by slow water flow velocity and shallow water depth (Dorioz et al., 1998; House and Warwick, 1998; Ensign et al., 2006). Furthermore, experimental data in a controlled environment confirmed that P concentration in stream water is not in equilibrium with bed sediment and that sorption phenomena predominate, at least during base flow conditions (M. Guidi, unpublished data). The net effect of the in-stream processes was approximated by a first-order decay

function, where the decrease in the mass stored in the reach is proportional to the reach length L (m) and a constant c (m^{-1}) representing the retention coefficient. According to Erturk et al. (2010), given the following:

$$dF / dL = -cF \quad [2]$$

the flux at node k after retention processes F_k^* (mg s^{-1}) is defined as

$$F_k^* = \sum_i \alpha F_i e^{cL_i} \quad [3]$$

The second new term Fw_k (mg s^{-1}) was introduced to include the mass entering with base flow along the reach; this term is positive because the ditches always drain the superficial aquifer. Therefore, the final outflow mass F_k^{**} (mg s^{-1}) at node k was obtained as follows:

$$F_k^{**} = F_k^* + Fw_k \quad [4]$$

The equation for outflow concentration becomes:

$$C_k^{**} = \frac{F_k^{**}}{Q_k} = \frac{\sum_i \left(C_i e^{-cL_i} \alpha Q_i\right) + Cw_k Qw_k}{\sum_i (\alpha Q_i) + Qw_k} \quad [5]$$

where Q_i is free water entering the ditch on its uphill section (m^3 s^{-1}), Qw_k the water table flow rate (m^3 s^{-1}), and C_i and Cw_k their respective concentrations (mg L^{-1}).

Therefore, solute concentrations were calculated along the drainage network, with the outflow concentration C_k^{**} and discharge Q_k considered, respectively, as the inflow concentration and discharge values for the downslope reaches. For initial reaches Q_i = 0, water and mass flow result only from water table contribution, except for point 2, where WWTP inputs had to be considered. In this case, Q_2 was set to a measured value of 0.01 m^3 s^{-1}. The following equation was used to estimate Qw_k according to Hooghoudt's assumptions (homogeneous area for hydraulic head, saturated hydraulic conductivity and drainage network geometry) and accepting steady-state flow conditions (Soutter et al., 2007):

$$Qw_k = q_c p L_k \quad [6]$$

where q_c is the specific discharge (m s^{-1}), p the length of the wetted section of the ditch, and L the reach length. Assuming that q_c and p are constant within a reference area, we can modify Eq. 6 as follows:

$$Qw_k = aL_k \quad [7]$$

where a is the "unit discharge inflow rate" (m^2 s^{-1}) defined per unit length. Introducing a "unit mass inflow rate" (parameter $b = Cw_k Qw_k / L_k$) defined as the flux of a compound entering a section of the ditch (mg m^{-1} s^{-1}), the term Fw_k can be calculated as $b_k L_k$. Combining Eq. 5 and Eq. 7 gives:

$$C_k^{**} = \frac{F_k^{**}}{Q_k} = \frac{\sum_i \left(C_i e^{-cL_i} \alpha Q_i\right) + b_k L_k}{\sum_i (\alpha Q_i) + aL_k} \quad [8]$$

In Table 1 are listed the input and output variables and parameters used in the simplified model.

To identify ditches and corresponding land areas involved in substantial nutrient losses and to define critical source areas, we

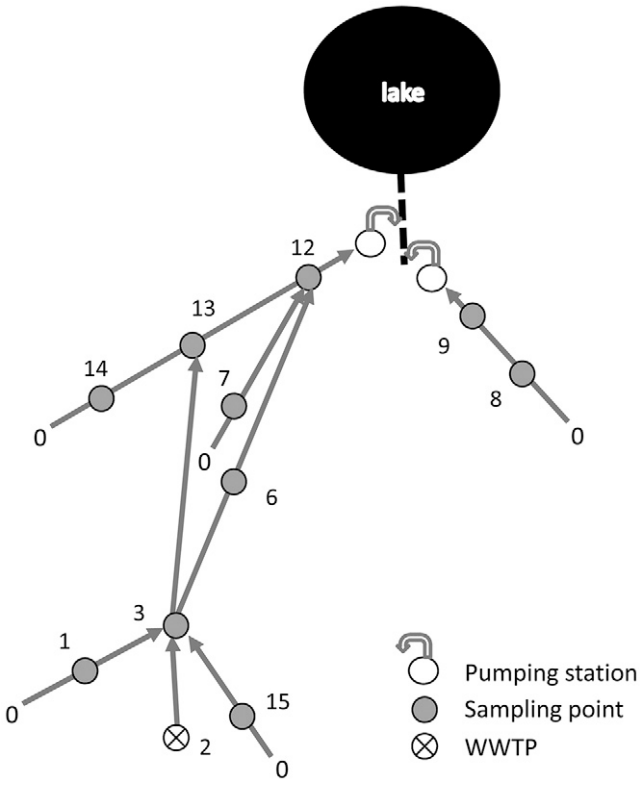


Fig. 2. Representation of node connectivity in the study area; 0 indicates initial reaches. WWTP, wastewater treatment plant.

Table 1. Input and output variables and parameters used in the simplified model.

Description	Name	Typology	Unit
Network geometry and stream direction	–	InV	–
Length of the reach	*L*	InV	m
Repartition coefficient	α	InV	–
Measured concentration of a given solute at the *k* node	*Ck*	InV	mg L^{-1}
Unit discharge inflow rate	*a*	P	m^2 s^{-1}
Unit mass inflow rate	*b*	P	mg m^{-1} s^{-1}
Retention coefficient	*c*	P	m^{-1}
Flux of a given solute at node *k* after retention processes	*Fk**	OutV	mg s^{-1}
Mass of a given solute entering with base flow along the *k* reach	*Fwk*	OutV	mg s^{-1}
Final outflow mass of a given solute at the *k* node	*Fk***	OutV	mg s^{-1}
Concentration of a given solute in the water table entering the *k* reach	*Cwk*	OutV	mg L^{-1}
Concentration of a given solute at the *k* node	*Ck***	OutV	mg L^{-1}
Water-table flow rate entering the *k* reach	*Qwk*	OutV	m^3 s^{-1}
Discharge at the *k* node	*Qk*	OutV	m^3 s^{-1}

considered two indicators (Table 3) that measure introduction or retention fluxes of chemical compounds per unit length—b and $\Delta F/L$, respectively—the latter defined as:

$$\frac{\Delta F}{L} = \frac{\sum_i \alpha F_i - \sum_i F_k^*}{\sum_i L_i} \quad [9]$$

where $F_k^* = F_k^{**} - b_k L_k$.

The simplified model contains the following assumptions: (i) The domain is homogeneous, isotropic, and continuous; (ii) baseflow conditions are verified (all sampling was performed outside heavy runoff events; base flow provides 80% of the water pumped into the lake each year) (Rossetto et al., 2010b); (iii) a first-order decay function approximates the in-stream processes of PO_4–P, TP, and NO_3–N; and (iv) the variability of F_w is negligible during each monitoring period, implying an inverse relationship between groundwater discharge and solute concentration.

The concentration dataset (C_k) was used to estimate parameters a, b, c and their 95% confidence limits by nonlinear least-squares fitting of Eq. 7, equaling C_k to C_k^{**}. All calculations were performed using Microsoft Excel Solver using a trial-and-error approach (Kemmer and Keller, 2010). In particular, parameter a was estimated by a mass-balance approach, in which water table flow rate was deduced from a subdataset limited to Cl and Na concentrations, assuming the conservation of these species ($c = 0$) and steady-state flow conditions. To include rainfall regimes and soil characteristics (Fig. 1c), different values of parameter a were calibrated on a monthly basis within each monitoring period and between the northern and southern zones of the study area. The values thus obtained were applied to estimate parameters b and c of nonconservative species (TP, PO_4–P, NO_3–N) for each reach. These parameters were allowed to vary spatially to reflect the effect of natural and anthropogenic changes in soil nutrient storage and release and temporally between the two monitoring periods to consider variations caused by seasonal and cultivation factors or hydrological and biophysical conditions of ditches. To increase the degrees of freedom of the fit, however, we assumed that b and c remained constant for all samplings within the same monitoring period.

The relatively simple model structure described by Eq. 8 can produce reliable results only if supported by a robust estimation procedure, mainly by estimating the order of magnitude of unknowns and comparing them with estimates obtained by complementary and independent analyses of the system. We thus introduced constraints to the adjustable parameters a and b by fixing minimum and maximum discharge rates (based on the expert knowledge of local reclamation district's staff) and considering that $b_{P-PO4} \le b_{TP}$.

Analytical Model Evaluation

The evaluation of model results was made by output variables validation and comparing the calibrated values of parameters with literature data and on-site measurements.

Among the output variables (Table 1), the validation involved only the estimated discharges (Q_k) because the concentrations (C_k^{**}) were obtained by the fitting procedures with the measured data set (C_k), whereas the fluxes (F_k^{**}) result from the product of C_k^{**} and Q_k (Eq. [8]).

Values of Q_k resulting from the parameter a calibration procedure were compared with values predicted by MODFLOW-2000 (Harbaugh et al., 2000) on a daily basis after calibration with the dataset of measured values of groundwater heads and water discharges collected over a year (Fig. 1a). Further details related to application of the hydrogeological model are found in Rossetto et al. (2010b).

The degree of agreement between the two series of Q_k values obtained by Eq. 8 and MODFLOW-2000 was evaluated by calculating model efficiency and coefficient of residual mass indices (Loague and Green, 1991) and relative error (RE).

To evaluate the extent to which retention values conformed to our knowledge of the system, we compared them with V_f, the mass transfer velocity obtained from the formula of Ensign et al. (2006): $V_f = cuh$, where c is the retention parameter (m^{-1}), u is the stream velocity (m s^{-1}), and h is the water column depth (m). Moreover, rates of soluble reactive P retention (rR_{SRP}), calculated as the median values of $\Delta F/L$ divided by the mean wetted width of reaches, were compared with rR_{SRP} values measured by tracer-injection experiments in other studies.

The estimated groundwater concentrations, calculated as b/a, were compared with the measurements made on-site by the regional environmental protection agency (ARPAT) in different monitoring campaigns.

Results and Discussion

Overview of Phosphorus and Nitrate Concentrations

Rainfall distribution displayed the pattern typical of a Mediterranean climate: P-I had lower rainfall (274 mm) than P-II (809 mm). Table 2 shows the mean measured concentrations of NO_3–N, PO_4–P, TP, and Cl for the two monitoring periods.

Spatial Pattern

Three patterns for PO_4–P and TP concentrations were observed: (i) they decreased in the downstream direction, with high concentrations immediately downstream from the WWTP (point 2), intermediate values at points 3 and 6, and lower values in the rest of the network; (ii) an inverse trend was found along the reach identified by points 14, 13, and 12, where concentrations of PO_4–P and TP progressively increased toward the outlet in both periods; and (iii) concentrations upstream from the peri-urban area (point 1) and in the Massaciuccoli subcatchment (points 8 and 9) were generally lower than those in ditches where WWTP discharges occurred.

Patterns of NO_3–N were equivalent to those of P, except for points 8 and 9, which showed the highest concentrations along the ditches draining cropland. This could be related to the higher mean N inputs: 128 kg ha^{-1} yr^{-1} (fertilizers and cattle slurry) vs. 107 kg ha^{-1} yr^{-1} (only fertilizers) for the Massaciuccoli and the Vecchiano subcatchments, respectively (Silvestri et al., in press). Chloride concentration patterns differed from those of nutrients, tending to increase toward the direction of the lake, which was subject to seawater recharge. Nutrient concentrations were lower in the lake than in the ditches network; immobilization in bed sediments, dilution, and storage in biomass are the main processes governing P status (Dorioz et al., 1998), whereas the reduction in NO_3–N concentrations was likely related to denitrification and algal uptake (Hellstrom, 1996; Ozen et al., 2010).

Temporal Pattern

Significantly higher values ($p < 0.01$) were observed in P-I for Cl concentrations but not for PO_4–P or TP concentrations. Dilution caused by differing rainfall amounts during the two monitoring periods probably played a significant role. For P species, dilution presumably was balanced by leaching from soil stocks, which is generally higher during late summer and early autumn due to an increase in the rate of soil organic matter mineralization (Di and Cameron, 2002; Sapek et al., 2007).

In contrast, nitrate concentrations were significantly higher ($p < 0.01$) in P-II than in P-I. Similar patterns have been observed in other cultivated peatlands of central Europe (van Beek et al., 2007; Tiemeyer et al., 2007). During P-I, the soil nitrate content likely was depleted by previous winter leaching, as frequently observed (Tournebize et al., 2008), despite peaks in nitrate concentrations observed in April and May (data not shown). Such high values likely resulted from summer crop N fertilization (maize and sunflower), as attested by the higher variability of NO_3–N concentrations during P-I (CV, 0.94 vs. 0.71 for P-II). The pattern described above suggests different dynamics for nitrates and P species; for the former, transport is mainly controlled by available stocks ("availability controlled"), whereas the latter greatly depends on solubility and sorption/desorption dynamics ("solubility controlled") (Bridgham et al., 2001; Zak et al., 2004).

Nutrients Transfer: In-Stream Retention and Diffuse Inputs

Phosphorus

For PO_4–P and TP, no significant differences were observed for b and $\Delta F/L$ between the two periods ($p > 0.05$). Phosphorus enrichment or retention is much more sensitive to spatial (differences between reaches) than temporal variations (differences between P-I and P-II). For PO_4–P, the highest b values were found for reaches located downstream from the WWTP (reaches 1–3, 2–3, 15–3, and 3–6), ranging from 0.007 to 0.017 and 0.003 to 0.009 mg s^{-1} m^{-1} for P-I and P-II, respectively. The area immediately upstream from the Vecchiano pumping station (reaches 13–12, 7–12, and 6–12) also had high values for parameter b (0.004 and 0.010 mg s^{-1} m^{-1} for P-II and P-I, respectively).

In the upper part of the subcatchment, PO_4–P enrichment along the drainage network seemed mainly due to high groundwater concentrations (ranging from 0.72 to 1.33 and from 0.21 to 0.46 mg L^{-1} for P-I and P-II, respectively). In contrast, high introduction rates near the pumping station likely were determined by high groundwater inflow rate.

Different dynamics for retention processes were observed for the two periods: in P-I the reduction in PO_4–P load ($\Delta F/L$) was higher for reaches 3–6 (0.012 mg s^{-1} m^{-1}), whereas in P-II a more homogeneous pattern was observed, with generally lower values. Reaches 13–12, 7–12, and 6–12 displayed a higher $\Delta F/L$ value (0.006 mg s^{-1} m^{-1}). Therefore, retention processes seemed more important in the middle part of the Vecchiano subcatchment (around point 6) and likely were related to lower stream velocity, promoting algal uptake, as confirmed by high chlorophyll levels (data not shown). Furthermore, cropping activities were assumed to provide few nutrients because no intensive farming was present in this area.

The estimated median and interquartile range of groundwater PO_4–P concentrations (0.11 and 0.03–0.33 mg L^{-1}, respectively) were similar to those measured in groundwater by the ARPAT in different monitoring campaigns (0.14 and 0.05–0.98 mg L^{-1}, respectively; ARPAT, unpublished data). This comparison partially confirms the accuracy of our estimates, corroborating their physical significance.

Regarding soluble reactive phosphorus retention, using mean values for velocity fields and water depths, the median values of V_f were 4.4 and 0.9 mm min^{-1} for P-I and P-II, respectively. These lay within the interquartile range reported by Ensign et al. (2006) of 1.0 to 7.2 mm min^{-1}. Moreover, our rR_{SRP} values (1.0 and 0.6 μg m^{-2} s^{-1} for P-I and P-II, respectively) were similar to those reported for natural base-flow conditions by Hill (1982) in agricultural streams (0.07–0.89 μg m^{-2} s^{-1}) and higher than

Table 2. Mean measured concentrations and coefficients of variation for each sampling site during the two sampling periods.

Site	NO_3–N		PO_4–P		TP†		Cl	
	P-I‡	P-II§	P-I	P-II	P-I	P-II	P-I	P-II
	mg L^{-1}							
1	0.42 (0.67)	3.22 (1.13)	0.005 (0.94)	0.051 (1.96)	0.21 (1.22)	0.52 (0.68)	77.0 (0.12)	224.9 (0.12)
15	3.38 (0.75)	6.30 (0.47)	0.080 (0.47)	0.224 (0.79)	0.26 (0.30)	0.28 (0.50)	214.1 (0.12)	72.8 (0.22)
2	5.36 (1.26)	13.53 (0.65)	1.835 (0.42)	2.139 (0.30)	3.23 (0.38)	3.72 (0.34)	296.8 (0.24)	263.6 (0.14)
3	2.27 (0.35)	6.15 (0.13)	0.753 (0.53)	0.232 (0.55)	1.00 (0.28)	0.28 (0.07)	217.5 (0.24)	130.4 (0.18)
6	2.55 (0.76)	5.75 (0.23)	0.329 (0.21)	0.184 (0.65)	0.58 (0.28)	0.37 (0.29)	167.0 (0.36)	134.4 (0.13)
7	3.44 (1.09)	5.87 (0.04)	0.243 (0.70)	0.042 (1.21)	0.67 (1.04)	0.50 (0.82)	402.4 (0.40)	373.3 (0.67)
14	3.25 (0.63)	2.87 (0.59)	0.026 (1.01)	0.020 (0.20)	0.16 (0.23)	0.11 (0.54)	526.5 (0.54)	281.4 (0.40)
13	3.74 (0.58)	4.72 (0.21)	0.074 (0.51)	0.032 (0.58)	0.25 (0.68)	0.30 (1.00)	740.6 (0.18)	459.5 (0.40)
12	3.05 (0.59)	6.21 (0.35)	0.127 (0.20)	0.048 (0.51)	0.33 (0.26)	0.25 (0.68)	734.6 (0.21)	445.7 (0.31)
8	9.09 (0.42)	9.22 (0.53)	0.017 (0.34)	0.001 (0.00)	0.16 (0.47)	0.08 (0.53)	467.3 (0.14)	333.7 (0.99)
9	7.27 (0.21)	6.79 (0.37)	0.092 (0.28)	0.017 (1.22)	0.29 (0.29)	0.10 (0.03)	507.2 (0.21)	353.7 (0.75)
Lake	2.09 (0.27)	2.37	0.003 (0.00)	0.001	0.03 (0.61)	0.04	1206.5 (0.03)	814.6 (0.16)
Mean	3.83	6.08	0.30	0.25	0.60		463.1	324.0
p Value	0.00544		0.07309		0.9560		0.00177	

† Total phosphorus.

‡ Period 1 (Feb.–May 2008).

§ Period 2 (Nov. 2008–Feb. 2009).

those reported by Munn and Meyer (1990) for forested streams (0.05–0.37 μg m^{-2} s^{-1}). They also were similar to the lower values reported by Macrae et al. (2003) (0.8 μg m^{-2} s^{-1}), but their study focused on first-order streams with a range of velocities an order of magnitude lower than those in this work.

Comparing TP with PO_4–P, the same b and $\Delta F/L$ dynamics were identified, although with higher values (up to double). For TP, the median value and interquartile range of measured groundwater concentrations (0.29 and 0.06–1.55 mg L^{-1}, respectively; ARPAT, unpublished data) were similar to our estimated values (0.38 and 0.18–0.75 mg L^{-1}, respectively). Estimated mean P retention rates were compared with rates observed by Gelbrecht et al. (2005) for ditches draining organic soils during the autumn-winter period (0.69 μg m^{-2} s^{-1}). Their value was lower than our estimates (1.18 and 0.95 μg m^{-2} s^{-1} for P-I and P-II, respectively) but was a net retention value, whereas our estimate was not. The higher retention values observed during P-I also agree with the observation that nutrient retention is an inverse function of stream velocity and consequently of the travel time through the network, as reported previously (Dodds et al., 2002; Seitzinger et al., 2002; Ensign et al., 2006).

Nitrate

Nitrate displayed different dynamics, with introduction remaining significantly higher in P-II ($p < 0.01$). Nitrate retention generally was lower than that for P, as illustrated by several negative values of $\Delta F/L$ obtained especially in reaches draining peat soils (points 13, 12, and 9) (Table 3). In these cases, c equaled 0 (no retention), and the negative $\Delta F/L$ value depended on the estimate of F_k^* (as $F_k^* = F_k^{**} - b_k L_k$) in Eq. [9], in which $\sum_i F_k^*$ is higher than $\sum_i \alpha F_i$.

The reasons for this inconsistency were the prevalence of release (e.g., nitrification of NH_4–N) over retention phenomena or the inadequacy of the *Fw* estimate (parameter *b*), for which the model provides a unique value within each period. The second hypothesis seems more likely because (i) nitrification should be much lower in such hydromorphic conditions and (ii) each period experienced high variability in NO_3–N concentrations. This indicates that the minimum monitoring interval should be shortened to reduce the length of the period in which b is assumed constant. Furthermore, observed ammonium dynamics highlighted the presence of non-negligible retention phenomena (data not shown) even though their spatial and temporal distribution does not seem to explain the observed inconsistencies.

In both periods, high values of introduction rates b were found in the Massaciuccoli subcatchment for reach 0–8 (b = 0.162 and 0.172 mg m^{-1} s^{-1} for P-I and P-II, respectively) and reach 8–9 (b = 0.111 and 0.148 mg m^{-1} s^{-1} for P-I and P-II, respectively). The highest value, however, was found for diffuse input from groundwater (b = 0.334 mg s^{-1} m^{-1}) only during P-II for the reaches connected to point 12 (6–12, 7–12, and 13–12). During P-I, application of inorganic N fertilizer could have increased groundwater NO_3–N concentration in the most intensively cultivated areas (northern part of both subcatchments). Conversely, during P-II, mineralization of soil organic-N, the lack of actively growing crops, and the higher inflow of groundwater prevail over fertilizers distribution for reaches 13–12, 0–8, and 8–9.

The mean prediction of groundwater nitrate concentration (6.90 mg L^{-1}) was higher than mean measured values (4.30 mg L^{-1}; ARPAT, unpublished data). The other distribution parameters showed even larger differences (median, 1.5 vs. 8.30 mg L^{-1}; interquartile range, 0.6–6.05 vs. 4.07–16.15 mg L^{-1} for measurements and predictions, respectively). Therefore, the reported b values can describe only average dynamics of nitrate inflow, concealing significant deviations.

Nutrients Transfer: Loads

Phosphorus

The estimated loads (F_k^{**} in Table 3) tended to increase in the downstream direction. Some exceptions were observed in both periods: In P-I the loads estimated at points 6 (PO_4–P and TP) and 13 (PO_4–P) tended to be lower than those at upstream points, and in P-II the same was observed for PO_4–P at points 12 and 13. These results likely were due to the retention processes observed in the corresponding reaches. The area upstream of points 6, 12, and 13, which constitutes the core of the Vecchiano subcatchment, seems to act as a buffer zone for the nutrient input of upstream points.

The increase in loads observed for the remaining points was related to discharge or to concentration increases; the latter was particularly high from point 8 to point 9 (outlet of the Massaciuccoli subcatchment), where PO_4–P concentrations increased from 0.017 to 0.092 mg L^{-1} during P-I and from 0.001 to 0.017 mg L^{-1} during P-II. This increase may have been due to seepage from the lake because the cropping systems upstream from each point were similar. In the Vecchiano subcatchment, although seepage from the lake occurred (points 14, 13, and 12), it seemed masked by higher discharges and upstream load retention.

Nitrate

The estimated loads (F_k^{**} in Table 3) increased in the downstream direction because, although discharges increased, concentrations remained constant along the network. This also confirms the low influence of nitrate retention processes. The high temporal variability observed during P-I (concentration peaks in April and May) apparently was due to fertilizer applications. During P-II, when no fertilizer was applied, nitrate loads were higher than in P-I. This suggests that peat soil mineralization constitutes an important contributor to surface water loads, as observed in cultivated peatlands of northern and central Europe (van Beek et al., 2004; Vermaat and Hellmann, 2010).

Overall nutrient dynamics in the study area displayed large variability. In particular, for P (soluble and total), three main areas were identified: (i) the southern zone of the Vecchiano subcatchment (points 1, 2, 15, and 3) where the large increases in nutrient loads were caused by high concentrations in groundwater or by high groundwater inflow rates, depending on the period (low- or high-rainfall period); (ii) the middle zone (points 6 and 7 and reaches 6–12 and 7–12), characterized by a reduction in P loads because of the prevailing retention processes; and (iii) the northern zone (points 8, 9, and 13 and reach 13–12), where most of the load was delivered from groundwater inflow in ditches due to seepage from the lake.

Therefore, options for mitigating these adverse effects should include pollutant sources and the factors influencing nutrient loading. In the northern area of the catchment, it is of utmost importance to limit nutrient release due to organic matter

mineralization (e.g., introduction of pluriannual crops such as short-rotation forestry, forage crops, or saturated pasture; adoption of soft-soil tillage techniques; rewetting low-lying areas) and to optimize the amount and timing of fertilization. In the southern area, where the main issue is the poor quality of wastewater, mitigation options should include improvement of WWTP efficiency and an increase in the buffering capability of the downstream drainage network (e.g., restoration of riparian vegetation, redesign of ditches geometry).

Relevance and Limitation of the Model

Estimated discharges (Q_k) obtained from the optimization of Eq. 8 increased from the uphill section (mean, 0.03 $m^3 s^{-1}$) to the drainage network outlets (mean, 0.75 $m^3 s^{-1}$), with higher mean values during the wetter period (0.20 and 0.28 $m^3 s^{-1}$ for P-I and P-II, respectively). Figure 3a compares discharges estimated with the simplified approach to MODFLOW predictions. These values showed a good level of agreement, as demonstrated by the model evaluation indices, in particular coefficient of residual

Table 3. Values of the upstream fluxes, total outflow fluxes, fluxes after retention processes, and parameters band for each node of the network.

	P-I†							P-II‡						
Node	F_k**§	F_i¶	*b*#	CI††	F_k*‡‡	ΔF	$\Delta F/L$§§	F_k**	F_i	*b*	CI	F_k*	ΔF	$\Delta F/L$
							PO_4–P							
1	0.1	–	0.000	–	0.0	–	–	1.4	–	0.001	–	0.1	–	–
15	2.6	–	0.000	–	1.2	–	–	18.8	–	0.003	–	7.3	–	–
2	18.1	–	–	–	–	–	–	21.0	–	–	–	–	–	–
3	64.6	20.8	0.017	0.006–0.027	31.2	0.2	0.000	42.3	41.3	0.003	0.000–0.010	35.3	6.0	0.002
6	14.7	23.2	0.007	0.000–0.009	2.5	20.8	0.012	29.2	15.2	0.009	0.005–0.015	13.4	1.8	0.001
7	7.1	–	0.005	–	0.5	–	–	1.1	–	0.001	–	0.1	–	–
14	4.1	–	0.001		0.6	–	–	4.4	–	0.000	–	1.5	–	–
13	32.5	45.4	0.003	0.001–0.005	2.8	42.6	0.004	20.3	31.4	0.002	0.001–0.003	1.9	29.5	0.003
12	76.5	54.3	0.010	0.003–0.016	21.2	33.1	0.006	41.4	50.7	0.004	0.000–0.008	18.7	32.0	0.006
8	1.1	–	0.000	–	0.1	–	–	0.1	–	0.000	–	0.0	–	–
9	11.6	1.1	0.003	0.002–0.005	2.1	–1.1	0.000	2.4	0.1	0.001	0.000–0.022	0.4	–0.3	0.000
							Total P							
1	4.0	–	0.001	–	1.2	–	–	13.7	–	0.006	–	1.5	–	–
15	7.9	–	0.001	–	4.0	–	–	23.5	–	0.003	–	9.5	–	–
	31.8	–	–	–	–	–	–	35.5	–	–	–	–	–	–
3	82.3	43.7	0.027	0.017–0.036	12.5	31.2	0.012	64.3	64.1	0.023	0.015–0.030	5.5	67.1	0.026
6	24.8	29.6	0.012	0.002–0.021	4.2	25.4	0.014	35.1	23.2	0.009	0.006–0.023	19.1	4.0	0.002
7	19.5	–	0.014	–	0.8	–	–	12.5	–	0.009	–	1.3	–	–
14	23.1	–	0.003	–	2.7	–	–	18.7	–	0.002	–	1.9	–	–
13	107.9	75.8	0.010	0.002–0.017	7.2	68.5	0.007	121.9	59.9	0.010	0.001–0.030	18.6	41.3	0.004
12	197.6	152.3	0.020	0.006–0.033	89.4	62.9	0.011	186.5	169.6	0.009	0.001–0.030	137.3	32.3	0.006
8	10.0	–	0.003	–	1.6	–	–	6.1	–	0.002	–	0.9	–	–
9	34.6	10.0	0.011	0.003–0.019	3.0	7.0	0.002	10.4	6.1	0.003	0.001–0.005	1.5	4.5	0.002
							NO_3–N							
1	8	–	0.003	–	1.4	–	–	96	–	0.043	–	10.5	–	–
15	72	–	0.012	–	0.0	–	–	528	–	0.089	–	157.4	–	–
	53	–	–	–	–	–	–	133	–	–	–	–	–	–
3	180	132	0.050	0.010–0.085	50.2	82.1	0.032	1042	756	0.144	0.010–0.038	668.1	88.3	0.034
6	156	65	0.074	0.045–0.095	26.7	38.1	0.022	536	375	0.065	0.001–0.220	422.6	–47.5	–0.027
7	75	–	0.025	–	42.7	–	–	179	–	0.113	–	32.0		
14	407	–	0.051	–	63.3	–	–	457	–	0.061		45.5	–	–
13	1492	523	0.074	0.016–0.010	710.9	–188.4	–0.018	2578	1124	0.145	0.100–0.340	1053.5	70.7	0.007
12	1919	1723	0.003	0.020–0.022	1902.0	–178.9	–0.033	5526	3293	0.336	0.100–0.800	3678.5	–385.7	–0.070
8	519	–	0.147	–	77.9	–	–	606	–	0.175		81.2	–	–
9	960	519	0.106	0.075–0.145	641.1	–121.9	–0.041	1063	606	0.146	0.070–0.300	625.8	–19.5	–0.006

† Period 1 (Feb.–May 2008).

‡ Period 2 (Nov. 2008–Feb. 2009).

§ Total outflow fluxes.

¶ Upstream fluxes.

Parameters brand $\Delta F/L = \left(\sum_i \alpha F_i - \sum_i F_k^*\right) / \sum_i L_i$.

†† Confidence interval (95%) for parameter *b*.

‡‡ Fluxes after retention processes.

§§ Reduction in PO_4–P load (see Eq. [9]).

mass (−0.025, optimal value = 0) and model efficiency (0.96, optimal value = 1). The slope of the regression line was equal to 1.02, and the regression was highly significant ($p < 0.001$). Residuals were randomly distributed (Fig. 3b). The REs generally ranged within ±20%. Point 8 had the highest RE (−150%) and generally the lower discharges (<0.06 $m^3 s^{-1}$), attributed to the higher level of uncertainty associated with estimates of *a* for the uphill portion of the network (Fig. 4). Except in November 2008, *a* values were greater in the northern zone of the study area (Fig. 4), consistent with its shallower water table.

The analytical model produces realistic predictions when some basic assumptions are met. These latter concern (i) the little influence of surface runoff in water discharge and nutrients transfer, (ii) the homogeneity of hydraulic and nutrients behavior between two nodes of interest, and (iii) steady conditions for water flow and nutrient transfer.

Such assumptions must be verified before any attempt to use the model in other ungauged surface drainage networks. Homogeneity must be established from the general knowledge of soil and practices conditions to evaluate the number and the distribution of the nodes enabled to represent the variability of the area. The steady-state condition is verified when the fluxes (Fw_k) and the retention and release phenomena may be considered as constant during the considered period (parameters *b* and *c* cannot change during time). Therefore, to verify this assumption, the period length must be adjusted to avoid significant temporal evolution of the parameters. As an example, the calibration results obtained here differ for each nutrients species, depending on whether the steady condition is verified or not. For P, concentrations appear mostly spatially controlled rather than temporally dependent. This reflects the combined effects of input location and the distribution of retention and immobilization sites along the watercourse. For such P dynamics, the assumptions we made to calibrate the model—spatially distributed retention and inputs considered constant for several months—appear acceptable. For nitrate, temporal differences in inputs led to high variability in concentration between sampling dates. Consequently, the nitrate calibration assumptions appear unrealistic; therefore, it is advisable to improve the description of the system under study, in particular by reducing the time between sampling dates as to shorten the length of the steady period.

The inverse use of the model, with *a*, *b*, and *c* as parameters and C_k as input variable, can be preferred, with the latter being more easily measured than the discharge and mass inflow rates or the retention coefficients. Moreover, its use is not suitable to predict the consequences of point and diffuse sources changes (cropping systems, rate of soil organic matter mineralization, wastewater management) in transient conditions and if users are concerned by critical events that may be responsible for rapid changes of discharge and nutrient transfer.

Conclusions

We developed a simplified analytical model that represents the main processes of introduction and retention of nutrients within a drainage network from chemical water concentrations, data which are easily accessible. The model provides an acceptable estimate of discharges and nutrient enrichment and retention phenomena. Discharge estimates are consistent with predictions of an independent hydrogeological model.

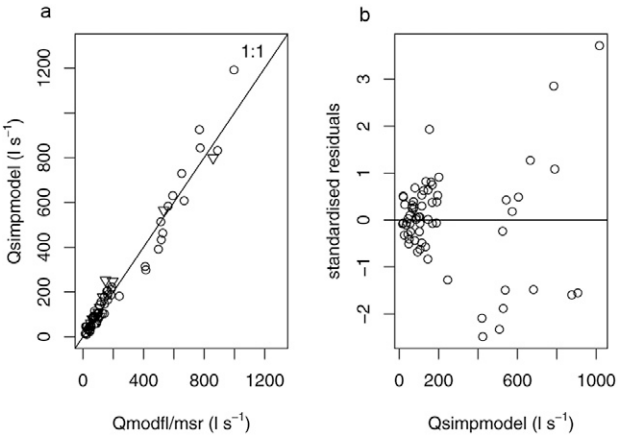


Fig. 3. (a) Comparison between the discharges (*Q*) estimated by the simplified model on the *y* axis and MODFLOW predictions (modfl, circles) and measured discharges (msr, triangles) on the *x* axis. (b) Standardized residuals of this regression.

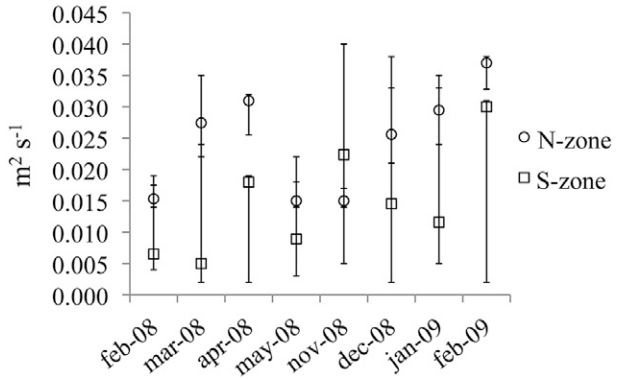


Fig. 4. Values of parameter *a* for the northern (N) and southern (S) zones of the study area. Error bars represent 95% confidence intervals. Null bars indicate that *a* values coincided with the adopted constraint.

To avoid over-fitting the model, given the limited number of observations available, we constrained model parameters and assumed that nutrient introduction and retention did not vary during the two periods studied. This simplification seems acceptable for P, whose estimates of introduction and retention parameters appeared consistent with the range of variability in measured concentrations and literature data, respectively. Conversely, the variability in nitrate concentration over time seems to affect model accuracy negatively and suggests the need for more intensive monitoring. In any case, the model's first-order decay functions can be modified to represent the differing influence of in-stream retention and release processes on P- and N-transport patterns. The model also separates the dynamics of different reaches of the drainage network, identifying zones that can be considered critical source areas emitting nutrients into the drainage network or as buffer zones where their transport is reduced. Although there may be reasons to prefer a more complex model over a simpler one, a relatively simple model of processes controlling the chemical composition of surface water can reliably reproduce part of the complexity of surface drainage networks and constitutes a surrogate solution when few data are available. In conclusion, this model appears suitable to explore the effects of nutrient release and retention phenomena in space and time when stable conditions at the chosen scales occur.

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